

WEEK 2 REPORT

DATA MODELING, DAX & DASHBOARD DEVELOPMENT

SUPERSTORE SALES ANALYSIS USING POWER BI

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Power BI • DAX • Data Visualization

1. Introduction

In Week 2, the cleaned Superstore dataset was used to develop an interactive Power BI dashboard. The main focus was on creating DAX measures, KPI cards and visualizations for sales, profit, category and regional analysis. Interactive features such as slicers, cross-filtering, tooltips and drill-down were also used.

2. DAX Measures and KPI Cards

A dedicated Measures table was used to keep the calculations organized. The dashboard includes measures for Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales and Profit Margin %. These measures were displayed through KPI cards to provide a quick summary of the dataset.

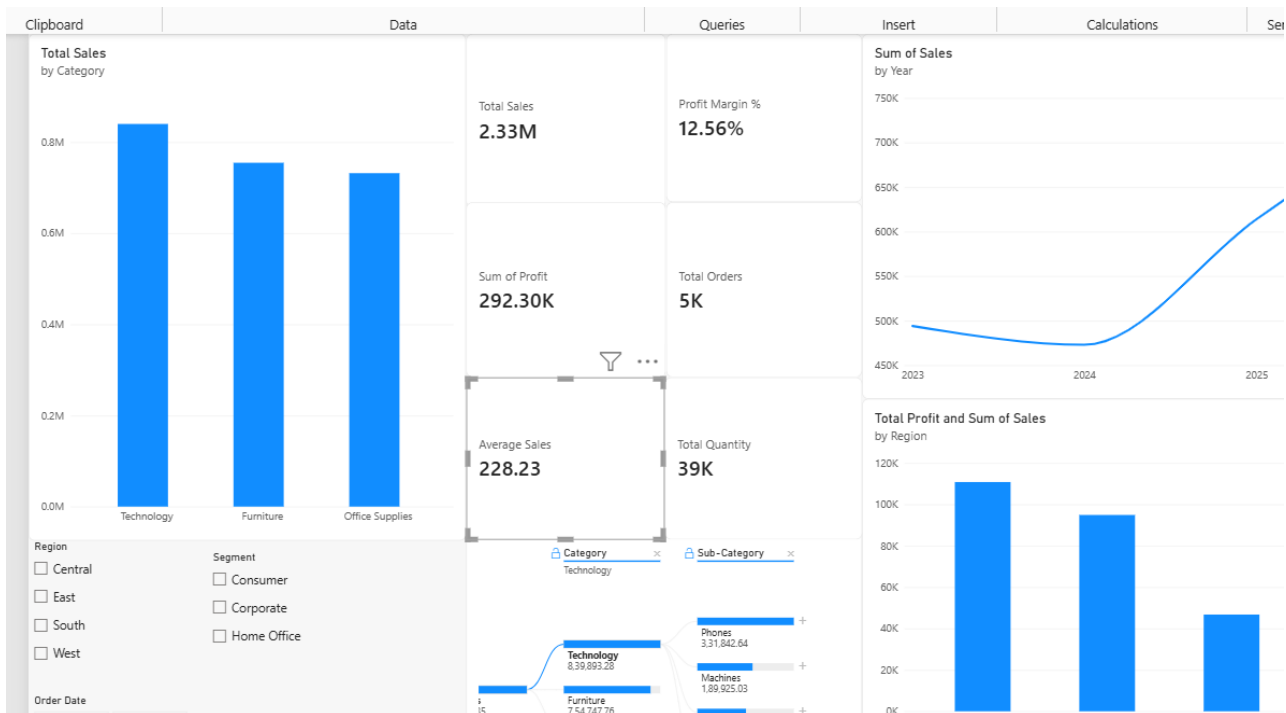


Figure 1. Main Power BI dashboard showing the KPI cards, category analysis and yearly sales trend.

3. Category Performance

A column chart was created to compare Total Sales across the three main categories: Technology, Furniture and Office Supplies. The Category and Sub-Category fields were also used to support detailed exploration of the sales values.

4. Sales Trend

A line chart was created using Year and Sales to show the change in sales over time. This provides a simple view of the yearly sales trend.

5. Drill-Down

Drill-down was applied to the Category and Sub-Category hierarchy. The chart can be viewed at the category level and then expanded to show the sub-categories within a selected category. Drill-up was used to return to the higher category level.

6. Regional Profitability

A regional chart was used to compare profitability across Central, East, South and West. This makes it easier to compare the contribution of each region to the overall performance.

7. Decomposition Tree

A decomposition tree was created for Total Sales. Category and Sub-Category were used as breakdown levels, allowing the total to be explored from the overall sales value into categories and individual sub-categories.

8. Stacked Bar Chart

A stacked bar chart was created with Region on the Y-axis, Sales on the X-axis and Segment as the legend. This visual compares sales across regions while also showing the Consumer, Corporate and Home Office segments.

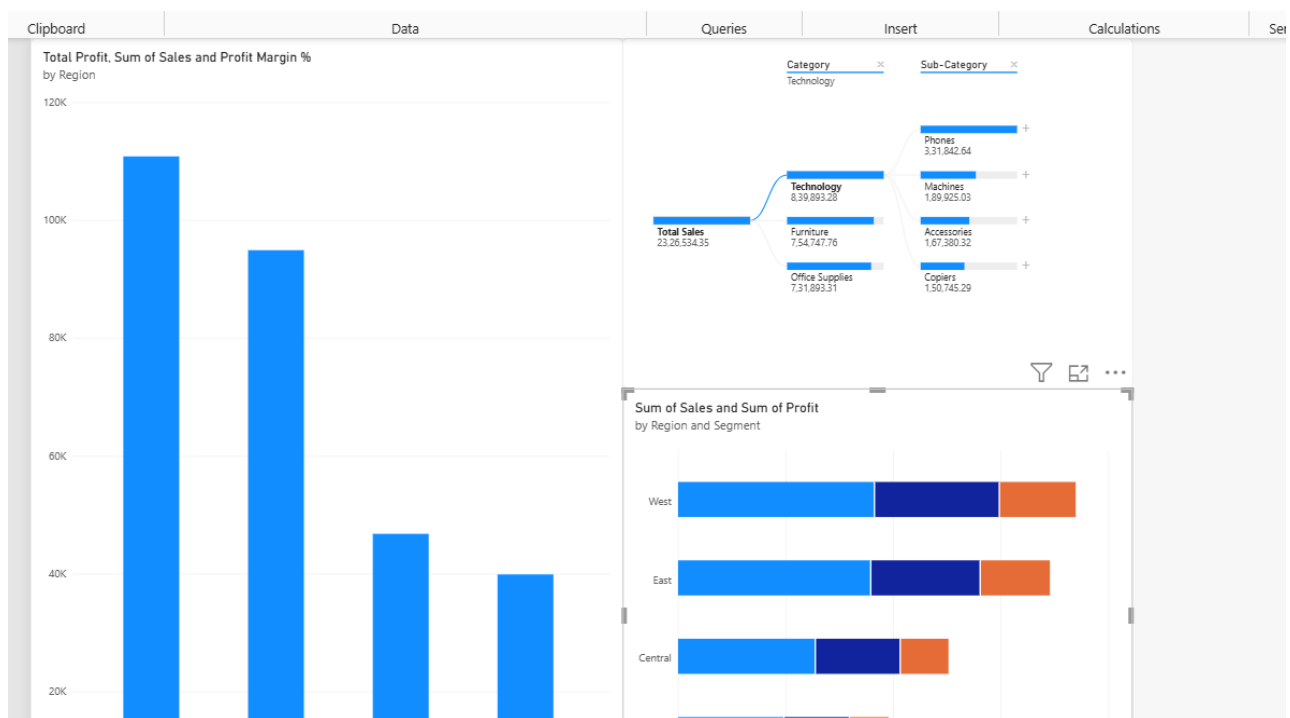


Figure 2. Regional profitability, decomposition tree and stacked bar chart used for analysis.

9. Slicers

Slicers were added for Region, Segment and Order Date. These controls allow the dashboard to be filtered interactively by selecting specific regions, customer segments or a date range.

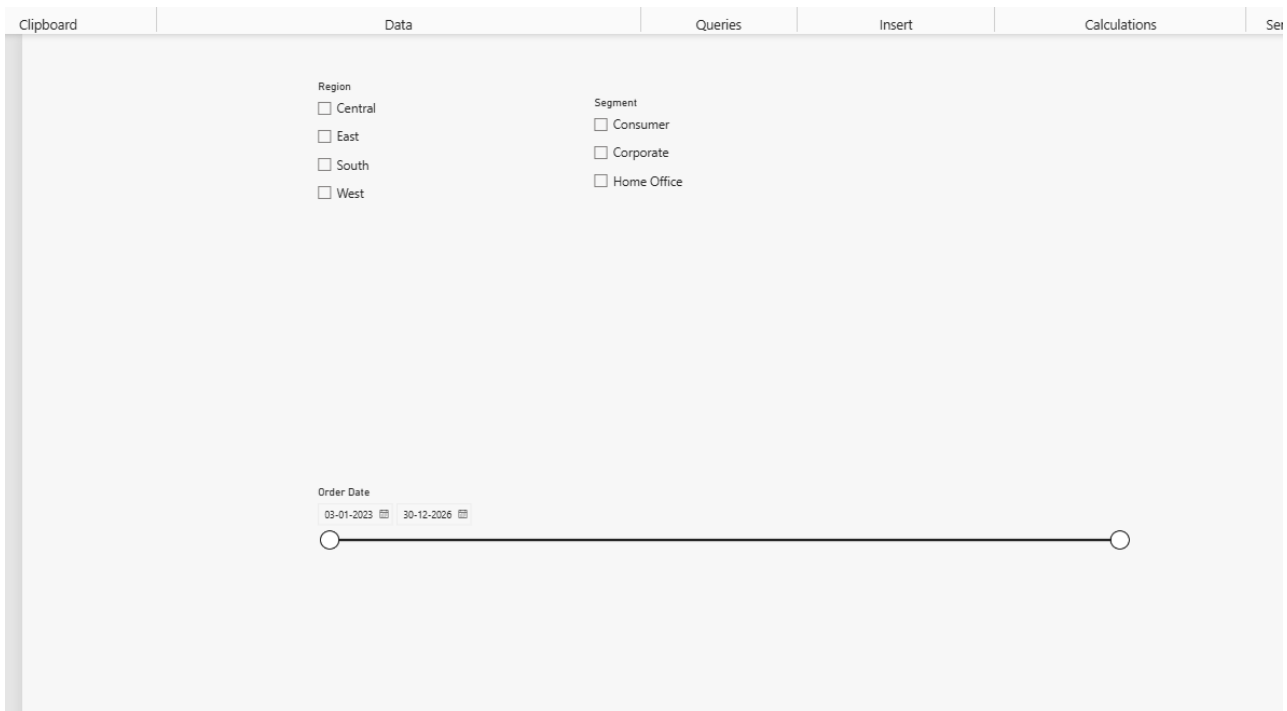


Figure 3. Slicers page showing Region, Segment and Order Date filters.

10. Cross-Filtering and Tooltips

The slicers and visuals were used together to test cross-filtering. Selecting a filter changes the displayed values in the related visuals. Additional information was also placed in the tooltip area of the relevant visual so that extra values can be viewed without adding another chart.

11. Dashboard Organization

The dashboard was arranged across multiple pages so that the visuals remain readable. The main page contains the summary KPIs and major visuals, while separate pages were used for slicers and supporting analysis.

12. Deliverables

- Dedicated Measures table for DAX calculations.
- Six KPI cards: Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales and Profit Margin %.
- Category and Sub-Category sales analysis with drill-down.
- Yearly sales trend line chart.
- Regional profitability analysis.
- Stacked bar chart using Region and Segment.
- Decomposition tree for Total Sales.
- Region, Segment and Order Date slicers.
- Cross-filtering and tooltip functionality.

13. Conclusion

Week 2 focused on turning the cleaned Superstore dataset into an interactive Power BI dashboard. The DAX measures provided reusable calculations, while the KPI cards and visualizations presented the main sales and profitability information. Drill-down, slicers, cross-filtering, tooltips and the

decomposition tree added interactivity and made the report easier to explore.